

Driver Fatigue Degree Detection Based on Eyes Opening and Closing Ratio

Qijie Zhao, Xianggang Wang, and Zhigao Lei

In the field of traffic safety, there is a growing need to measure driver fatigue degree. Fatigue is a complex physiological and psychological state, and the eye status can directly reflect the fatigue situation. It is important to define the appropriate metrics of the eye status and construct a detection model to identify the fatigue degree. This paper proposes a new fatigue degree detection method based on the eye opening and closing ratio and Gaussian Mixture Model. The discussion is supplemented by a brief case study highlighting the feasibility of the proposed approach.

Technical Context

Capturing and analyzing physiological signals of drivers to detect their fatigue level in the automobile cockpit is an important way to ensure driving safety today. A driver's fatigue is indeed related to many different factors and physiological metrics, and the physiological signals of drivers can be captured with invasive or non-invasive sensors. The invasive sensors can directly contact with drivers and accurately monitor various physiological metrics, including heart rate, skin electrical activity, etc. This kind of method is beneficial for obtaining physiological signal information, but wearing monitoring devices is not suitable for driving. On the contrary, non-invasive sensors, such as those in cameras, have the advantages of being non-invasive and non-contact with drivers. Cameras use image processing technology to analyze eye morphology features and eye movement behavior to complete driver fatigue detection. The eye status can directly reflect the fatigue situation, and the eye metrics are more direct and important for fatigue degree identification. Therefore, it is necessary to design an efficient and convenient fatigue detection system in light of the metrics of the eye status. This paper designs a reliable and efficient fatigue measurement system with camera sensors, cutting-edge image processing methods, and machine learning algorithms.

The RGB cameras can obtain texture clear images under sufficient lighting conditions and can extract facial feature points and eye metrics through RGB images [1]. However, it is a big challenge to measure fatigue parameters by separating the eye and the pupil area for the limitation of external

factors, such as light, face occlusion, and sunglasses. There are lots of researchers who have proposed to use near-infrared cameras [2],[3] to collect images and get better results. This paper designed a customized infrared module (Fig. 1) to collect a driver's facial image, and the combination of an infrared auxiliary light source and filter is implemented for the camera module to present clear images even in the strong daylight environment, which meets the requirements of vehicle working environments.

To identify the fatigue degree with the metrics of eye status, the Region of Interest (ROI) feature extraction of eyes is particularly important, because the extraction results of iris and pupil parameters will directly affect the detection accuracy. Traditional methods mostly rely on detecting the position of the pupil and accurately locating the iris region, such as in template matching [4], Hough transform [5], threshold segmentation, etc. In recent years, deep learning has been widely used in iris segmentation, including techniques such as CNN [6], U-Net [7], SegNet [8], etc. These methods perform better in low-quality iris images compared with traditional ways. Because of the large distance between the infrared acquisition module and the driver, the obtained pupil resolution of the driver's eyes is relatively low. Considering that deep learning performs well in low resolution scenes, this paper uses this method to study the iris segmentation problem in fatigue detection.

In addition, some researchers have also work based on multi-sensor fusion. For example, research has involved: fusing drive status and vehicle behavior characteristics through collecting partial facial and steering related parameters [9]; fusing driver state and physiological signal features by acquiring facial feature parameters and heart rate parameters [10]; and fusing the driver status parameters, vehicle behavior parameters, and physiological signal [11]. The method of multiple sensing fusion can comprehensively judge driver fatigue with the redundant information, and simultaneously, this kind of method also hides the mutual relationship between separate factors (such as metrics of eye status) and the fatigue. From the aspect of fatigue degree identification, revealing the influence of eye status

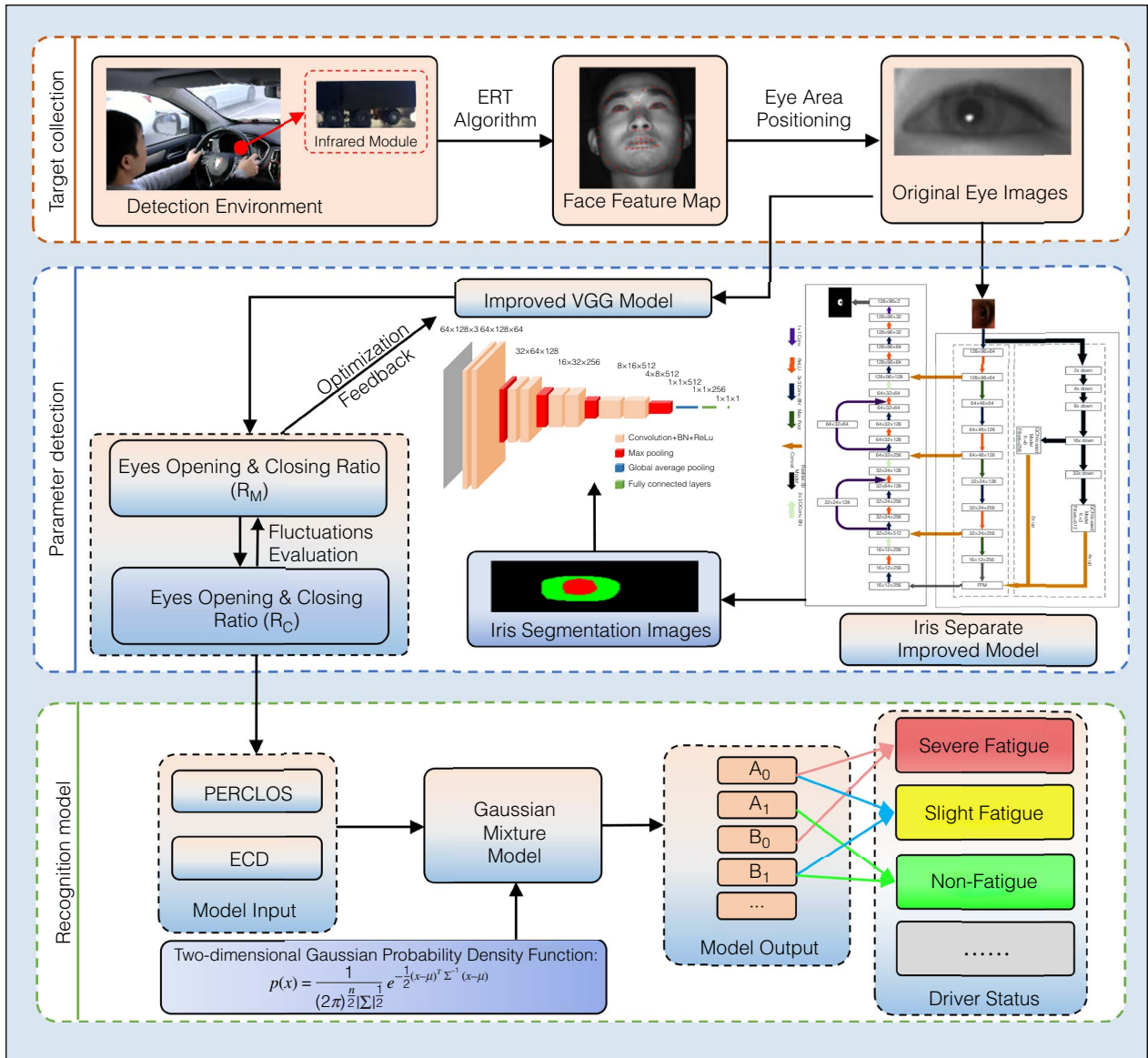


Fig. 1. Driver fatigue degrees detection framework.

parameters on fatigue degree is very important. Therefore, this paper focuses on the parameter detection of driver's eye status to reveal the relationship between the metrics of eye status and fatigue degree, which aims to better improve the robustness of driver fatigue recognition for multi-metrics fusion.

Driver Fatigue Degrees Detection Framework

Previous studies have found that identifying the iris and pupil of the driver's eyes plays a key role in the detection of the eyes opening and closing ratio value defined as EOCR [12]. To more accurately assess the fatigue level of drivers, iris segmentation and EOCR value detection algorithms are applied to the infrared images of drivers' eyes in driving environments to: achieve accurate iris segmentation; extract eye parameters; and detect the EOCR value of the driver, ultimately detecting

different levels of driver fatigue. The overall framework is divided into three parts: image acquisition, the EOCR detection, and fatigue degrees subdivision.

Establishing an improved iris segmentation network model is based on U-Net and BiSeNet, to minimize the information loss caused by direct up-sampling and obtain rich spatial and contextual information. The original image of the eyes is input into the model to obtain feature maps, and the model finally outputs the iris separation image. This image can be used to calculate the pixel scale to generate the EOCR value R_C for the driver. To better preserve information, the feature map output by the iris segmentation network model is connected to the original image, and the resulting Concat layer is input into the improved VGG module, which then outputs the generated EOCR value R_M for the driver. The R_C and R_M values obtained by this method are evaluated for fluctuations using

standard deviation and are superior to the results obtained by training a simple VGG network in terms of accuracy and stability.

The R_M values are processed with the parameter detection method to calculate Percentage of Eyelid Closure Over the Pupil Over Time (PERCLOS) and Eye Closure Duration (ECD) parameters that reflect the duration of each eye closure. These parameters are input into a Gaussian Mixture Model and output through different standards to obtain indicators such as A_0 , A_1 , B_0 , B_1 , etc. By using these indicators, the driver fatigue is identified at different levels, divided in this article into three states: severe fatigue, slight fatigue, and non-fatigue, which can be further subdivided according to requirements.

Iris Segmentation

The U-Net network architecture introduces an Encoder-Decoder structure. This structure concatenates shallow features with deep features in the depth direction of the decoding section. Then it constructs a new feature map containing deep contextual information and shallow positional information through the use of four down-sampling and four up-sampling. These features are fused through convolutional layers. The BiSeNet network structure consists of three main parts: Spatial Path, Context Path, and Feature Fusion Module (FFM). The network extracts location information through a shallow network of simple spatial path modules and then uses complex networks with context path modules to obtain deep contextual information. The feature fusion module is used to fuse and solve the problem of coexistence of these two types of information.

The most important problem faced by semantic segmentation models in the establishment process is the contradiction between spatial information located in shallow networks and contextual information located in deep layers. The advantage of BiSeNet network lies in its independent spatial feature module and contextual feature module, which respectively extract spatial and contextual information and fuse them in the feature fusion module. The disadvantage is that using 8x up-sampling directly results in partial loss of detail information during the up-sampling process. On the contrary, the information extraction of the U-Net network is relatively simple, and the rigorous encoding decoding structure preserves all information as much as possible. Taking into account the advantages of both networks, the main structure of BiSeNet is used as the encoding part, and a global convolution module is designed in combination with large kernel convolution, which can obtain contextual information to a greater extent. On the other hand, by designing the decoding part based on the U-Net network, the feature map of the spatial extraction module is overlaid with the feature map of the decoding part, which can obtain rich spatial and contextual information while minimizing the information loss caused by direct up-sampling. Additionally, a residual module is introduced to make the algorithm easier to converge. The loss function used in training the iris segmentation network is the average binary cross entropy, as in:

$$\text{loss}_1 = \frac{1}{M \times N \times B} \sum_{k=1}^B \sum_{j=1}^N \sum_{i=1}^M t_{ij} \log(p_{ij}) - (1 - t_{ij}) \log(1 - p_{ij}) \quad (1)$$

where M and N represent the length and width of the output image, respectively, B represents the batch size, and p_{ij} and t_{ij} represent the calculated values of the softmax function corresponding to the predicted image and the target image, respectively.

Improved VGG Model

The VGG network structure is very simple, consisting of a series of convolution layers and pooling layers of the same size stacked together. It achieves excellent performance in image processing tasks and makes it easy to transfer learning for other visual tasks. Therefore, this model is used as a feature extraction network for eye images and iris segmentation feature maps to obtain the EOVR values. Due to the large number of convolutional layers in the model, there are issues such as high parameter count, high computational complexity, and overfitting. Therefore, a variant of the VGG model was obtained with two main improvements (Fig. 2). Firstly, BN and ReLU layers were added after each convolutional layer to prevent overfitting problems. Secondly, the fully connected layer of the first one was replaced with a global average pooling layer to reduce computational parameters and complexity.

For the purpose of better information retention, the feature map obtained from the improved iris segmentation network model and the original image are concatenated, and the new Concat layer is obtained as input after processing, with the output being the driver's EOVR value R_M . This value and the R_C value obtained above are evaluated for fluctuations, and the model is further improved by the corresponding optimization feedback mechanism. The loss function of the model is defined as:

$$\text{loss} = \text{loss}_1 + \alpha |x - x^*| \quad (2)$$

where x represents the trained open close size, x^* represents the theoretical open close size value, α represents the weight, and loss_1 represents the loss function of the first part.

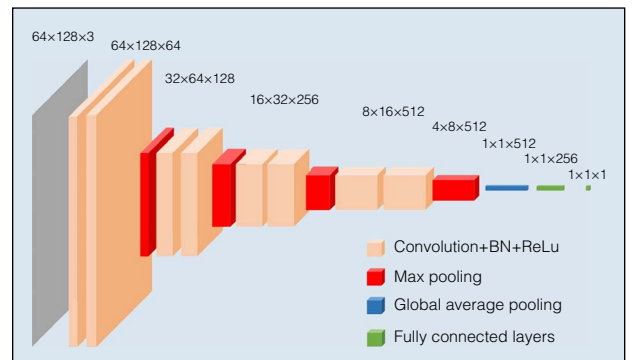


Fig. 2. Improved VGG network structure diagram.

By using this loss function, the loss of both the first and second segments of the network can be optimized simultaneously to achieve end-to-end performance. High quality iris segmentation images can be obtained while obtaining the opening and closing sizes.

Gaussian Mixture Model

Currently, the PERCLOS parameter is the most used and authoritative parameter in driver fatigue detection. Compared with the PERCLOS parameter for the total time of eye closure over a period of time, the ECD parameter focuses more on the duration of each eye closure. Fatigue state is determined by calculating the ratio of the number of frames to the number of closed eyes over a period of time. The expressions of PERCLOS [13] and ECD [14] parameters are:

$$PERCLOS = \frac{\sum_{i=1}^n Close[i]}{n} \times 100(\%) \quad (3)$$

$$ECD = \frac{\sum_{i=1}^p Duration[p]}{p} \quad (4)$$

Table 1 lists the parameters for the two different fatigue judgment indicators in PERCLOS parameters-EM and P80 standards.

The definition diagram of EOCR [12] proposed is shown in Fig. 3. The solid shaded part of the figure represents the closed area of the human eye as defined by the EM standard, and the dashed shaded part as defined by the P80 standard. The higher value of PERCLOS indicates a greater proportion of the total duration of eye closure and a greater probability of fatigue. Since fatigue is a complex physiological phenomenon that is influenced by multiple metrics, using a single parameter to characterize fatigue status is not reliable. Therefore, using a Gaussian Mixture Model combined with PERCLOS and ECD parameters can improve the robustness of fatigue detection.

The model is a probabilistic model, which is widely used for modeling complex multidimensional distributions. In this paper, only a two-dimensional Gaussian Mixture Model is used. It can also be combined with other kinds of fatigue parameters for multidimensional fusion, which certainly has some challenges for the complexity of the algorithm and the degree of the fitting. The two-dimensional Gaussian probability density function is:

$$p(x) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \quad (5)$$

Table 1 – Eye opening and closing status judgment indicators				
Judgment Indicators	EM Standard		P80 Standard	
EOCR	≥50%	<50%	≥80%	<80%
Eye status	Open	Closed	Open	Closed

where μ represents the n-dimensional mean vector. n is equal to 2 and represents the two parameter values of PERCLOS and ECD, respectively. The mean vector μ and the covariance matrix Σ determine the entire Gaussian distribution, and the expression of the Gaussian mixture distribution is:

$$p_M(x) = \sum_{i=1}^k \alpha_i \cdot p(x | \mu_i, \Sigma_i) \quad (6)$$

where k represents the distribution consisting of k mixture components. k is equal to 2, representing two driving states of fatigue or not, α_i represents the mixing coefficient, and $\alpha_i > 0$.

According to Bayes' theorem, for the specific state z_i to be measured, the posterior probability is calculated to verify which driving state it is. The specific expression is:

$$p(z_j = 1 | x_j) = \frac{\alpha_i \cdot p(x_j | \mu_i, \Sigma_i)}{\sum_{l=1}^k \alpha_l \cdot p(x_j | \mu_l, \Sigma_l)} \quad (7)$$

For the convenience of presentation, $p_M(z_j = i | x_j)$ is written as γ_{ji} ($i = 1, 2, \dots, k$) in this paper. The posterior probability of the fatigue state is calculated by the Gaussian Mixture Model, and the overall fatigue state is determined by comparing the values of the posterior probabilities. The parameters which are solved iteratively using the EM algorithm with the iterative formulas are:

$$\mu'_i = \frac{\sum_{j=1}^m \gamma_{ji} x_j}{\sum_{j=1}^m \gamma_{ji}} \quad (8)$$

$$\Sigma'_i = \frac{\sum_{j=1}^m \gamma_{ji} (x_j - \mu'_i)(x_j - \mu'_i)^T}{\sum_{j=1}^m \gamma_{ji}} \quad (9)$$

$$\alpha'_i = \frac{\sum_{j=1}^m \gamma_{ji}}{m} \quad (10)$$

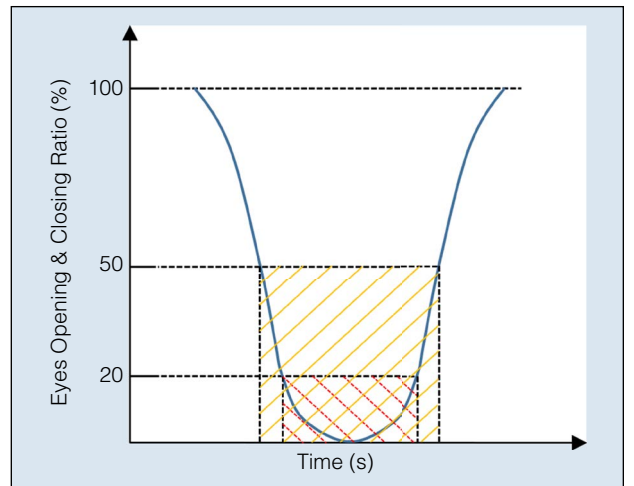


Fig. 3. EOCR definition chart.

Short Case Study

A brief report on the application case of the driver fatigue detection framework in real vehicle testing environment and the specific environment is shown in Fig. 4a. The mounting position of the camera plays a very important role, considering not only the integrity of the captured images and the accuracy of image detection, but also the possibility of mounting in the actual vehicle. From the perspective of installation, there is more space on the upper part of the instrument panel, considered a more reasonable installation position (Fig. 4b), so the design is coordinated and does not block the panel.

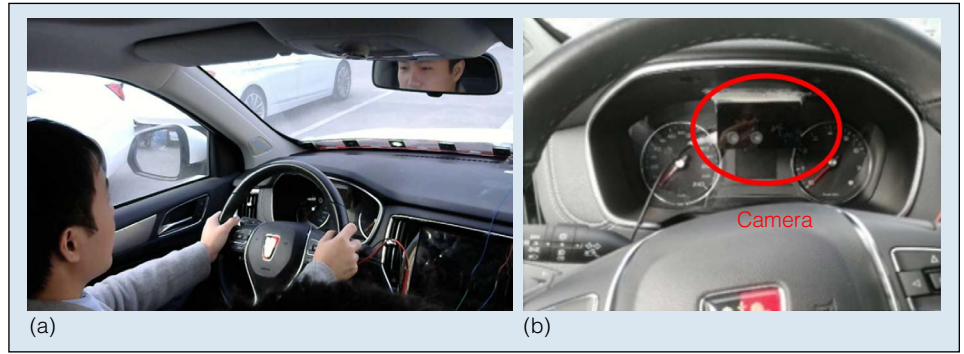


Fig. 4. The testing environment. (a) The detection environment; (b) The installation of camera.

Iris Segmentation Model Validation

This article validates the effectiveness of the proposed iris segmentation model by comparing it with other methods on the publicly available dataset CASIA interval using recall (R), accuracy (P), and F_1 -scores to evaluate the model. The recall rate represents the model's ability to recognize positive samples, the accuracy represents the model's ability to recognize negative samples, and the F_1 -score represents the combined ability of both. The comparison results of the model are listed in Table 2, where μ and σ represent the mean and standard deviation of R , P , and F_1 , which can reflect the accuracy and stability of the model. The results indicate that the proposed semantic segmentation model performs outstandingly in terms of accuracy and stability.

Improved VGG Model Validation

To verify the effectiveness of the improved VGG model, facial images of six drivers were collected in three different lighting environments based on the constructed real vehicle testing environment: morning (10:00-11:00), afternoon (15:00-16:00), and evening (19:00-20:00). We established a sample database including iris and pupil information, with a total of 1126 images, 907 for the training set, 108 for the validation set, and 111 for the testing set. After training, the performance of VGG and the

improved VGG model were tested, using standard deviation s for fluctuation evaluation:

$$s = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - x_i^*)^2} \quad (11)$$

where x_i represents the network output value of the detected EOVR, and x_i^* represents the actual opening and closing size. The smaller the value of s , the smaller the deviation between the predicted value and the actual value of each frame.

The measurement standard deviation of the VGG model in the established dataset is 0.0829, while the improved VGG model corresponds to a measurement standard deviation of 0.0544. The results indicate that the improved VGG method outperforms the original VGG model in terms of accuracy and stability.

Gaussian Mixture Model Validation

Based on the constructed testing environment and customized infrared camera, three videos were collected from six different volunteers in non-fatigue, slight fatigue, and severe fatigue states. In addition, the shooting time was divided into three periods, starting from 10:00 to 11:00, 15:00 to 16:00, and starting at 19:00. Each video clip consists of 200 frames of images, with a total duration of 1 minute and 20 seconds. Examples of images from the testing are shown in Fig. 5.

The designed model considers the situation where both eyes are in different states, detects the EOVR values of both eyes separately, and analyzes the differences between the same person on the same image frame. The calculation expression that defines the average disparity value of two-eye EOVR is:

Table 2 – Comparison results of iris segmentation methods						
Method	R		P		F_1	
	μ	σ	μ	σ	μ	σ
GST	85.19	18	89.91	7.37	86.16	11.53
IFFP	91.74	14.74	83.5	14.26	86.86	13.27
IrisSeg	94.26	4.18	92.15	3.34	93.1	2.65
IrisDenseNet	97.50	2.12	98.10	1.07	97.58	0.99
Ours	97.07	1.68	94.58	5.15	95.72	2.99

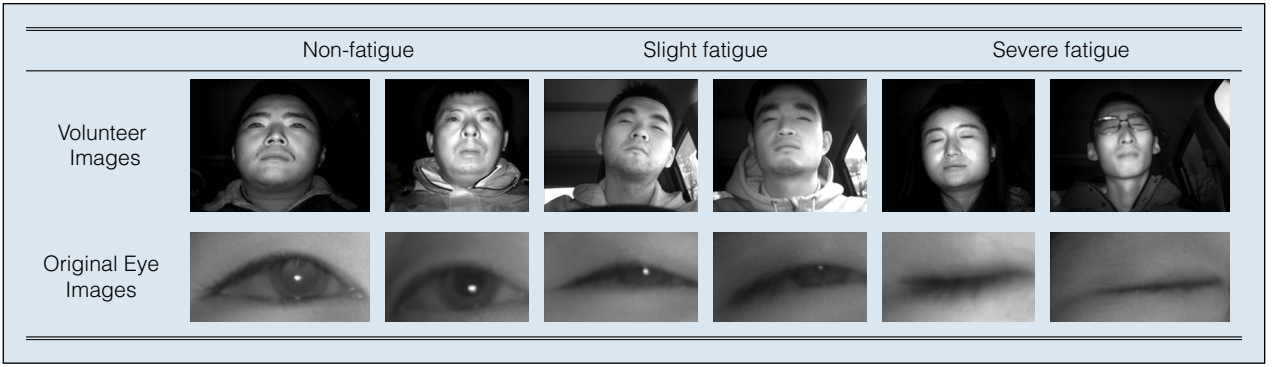


Fig. 5. Example images of volunteers.

Video number	Average Gap $\bar{O}$	Video number	Average Gap $\bar{O}$
01	0.0514	10	0.0938
02	0.0777	11	0.1084
03	0.1474	12	0.1626
04	0.0876	13	0.0514
05	0.1081	14	0.0480
06	0.0574	15	0.1222
07	0.0699	16	0.0694
08	0.0765	17	0.0863
09	0.0936	18	0.1037

$$\bar{O} = \frac{\sum_{i=0}^{frame} |O_i^{left} - O_i^{right}|}{frame} \quad (12)$$

where O_i^{left} is the EOVR value of driver's left eye at frame i , while O_i^{right} is for the right one, and $frame$ is the number of frames of the video. The specific results are listed in Table 3.

Two different judgments based on PERCLOS are used to determine the degree of fatigue. The biggest advantage of using the EOVR value is that the degree of fatigue can be analyzed, which is more in accordance with the initial principle of PERCLOS parameters. The expression for calculating the overall detection variance S_W value of the driver's EOVR for the video segment is:

$$S_W = \frac{\sum_{i=0}^{frame} (R_M - R_C)^2}{frame} \quad (13)$$

The specific EOVR value variance and fatigue judgment results of 18 video segments are listed in Table 4. We recorded the overall detection variance S_2W between the EOVR network output values and pixel calculation values of different volunteers at different periods, as well as the states derived from the Gaussian Mixture Model output based on the EM and P80 standards. These states are mapped to parameters such as A_0 , A_1 , B_0 , and B_1 , which are then subdivided into the

different fatigue degrees (non-fatigue, slight fatigue, and severe fatigue) and compared with the actual recorded driver states.

The real-time detection effect during the experiment is dynamically reflected by calculating the real-time driver's EOVR detection deviation S_R value. The $S_R[i]$ value of video i is:

$$S_R[i] = (R_M[i] - R_C[i])^2 \quad (14)$$

The box-plots in Fig. 6 show that the experimental deviation S_R of the upper and lower quartiles are between 5% and 10%, performing a good detection result. The overall detection variance value of the driver's EOVR is not large, which indicates that the accuracy of the driver's EOVR is relatively stable and the fatigue state is also judged more accurately. Data listed in Table 3 show that in video clips representing non fatigue and severe fatigue, the difference in average EOVR values between the left and right eyes of the driver is very small, with an average value of 7.73%. The value indicating mild fatigue is about 11.44%. This is mainly due to the slight difference in the speed at which people close their left and right eyes due to personal habits and other reasons. From the overall detection deviation value S_W for the EOVR, the vast majority of values are within 0.1 and more than 50% are within 0.05. In fact, drivers are more prone to fatigue in complex lighting environments such as noon or evening. From the overall detection deviation and real-time video results, good results can be obtained in both complex scenarios.

Conclusion

The driver fatigue detection model proposed in this article has the advantage of non-invasiveness compared to fatigue detection methods based on physiological signals [10]. Compared with fatigue detection methods based on vehicle operating status [11], the proposed model has stronger anti-interference ability. In computer vision methods, current research methods [1],[9] mostly rely on the distance transformation information of human eyes to determine fatigue, thus ignoring the subdivision judgment of fatigue level. In contrast, this article extracts characteristic parameters such as iris and pupil of the eyes, which can reflect the degree of eye opening, thereby subdividing and detecting the driver's fatigue level. It is worth noting

Table 4 – Driver opening and closing state combined with fatigue detection results

Video number	Volunteers	Time	S_w	EM Standard	P80 Standard	Recognition results (objective)	Real fatigue state (subjective)
01	1	10:00~11:00	0.0242	non-fatigue	non-fatigue	non-fatigue	non-fatigue
02	1	10:00~11:00	0.0459	fatigue	fatigue	severe fatigue	severe fatigue
03	1	10:00~11:00	0.0424	fatigue	non-fatigue	slight fatigue	slight fatigue
04	2	10:00~11:00	0.0471	non-fatigue	non-fatigue	non-fatigue	non-fatigue
05	2	10:00~11:00	0.0844	fatigue	fatigue	severe fatigue	severe fatigue
06	2	10:00~11:00	0.0612	fatigue	non-fatigue	slight fatigue	slight fatigue
07	3	19:00~20:00	0.0498	non-fatigue	non-fatigue	non-fatigue	non-fatigue
08	3	19:00~20:00	0.0771	fatigue	fatigue	severe fatigue	severe fatigue
09	3	19:00~20:00	0.0506	fatigue	non-fatigue	slight fatigue	slight fatigue
10	4	15:00~16:00	0.0437	non-fatigue	non-fatigue	non-fatigue	non-fatigue
11	4	15:00~16:00	0.0647	fatigue	fatigue	severe fatigue	severe fatigue
12	4	15:00~16:00	0.1114	fatigue	fatigue	severe fatigue	slight fatigue
13	5	19:00~20:00	0.0325	non-fatigue	non-fatigue	non-fatigue	non-fatigue
14	5	19:00~20:00	0.0375	fatigue	fatigue	severe fatigue	severe fatigue
15	5	19:00~20:00	0.0665	fatigue	non-fatigue	slight fatigue	slight fatigue
16	6	15:00~16:00	0.0420	non-fatigue	non-fatigue	non-fatigue	non-fatigue
17	6	15:00~16:00	0.0706	fatigue	fatigue	severe fatigue	severe fatigue
18	6	15:00~16:00	0.0702	fatigue	non-fatigue	slight fatigue	slight fatigue

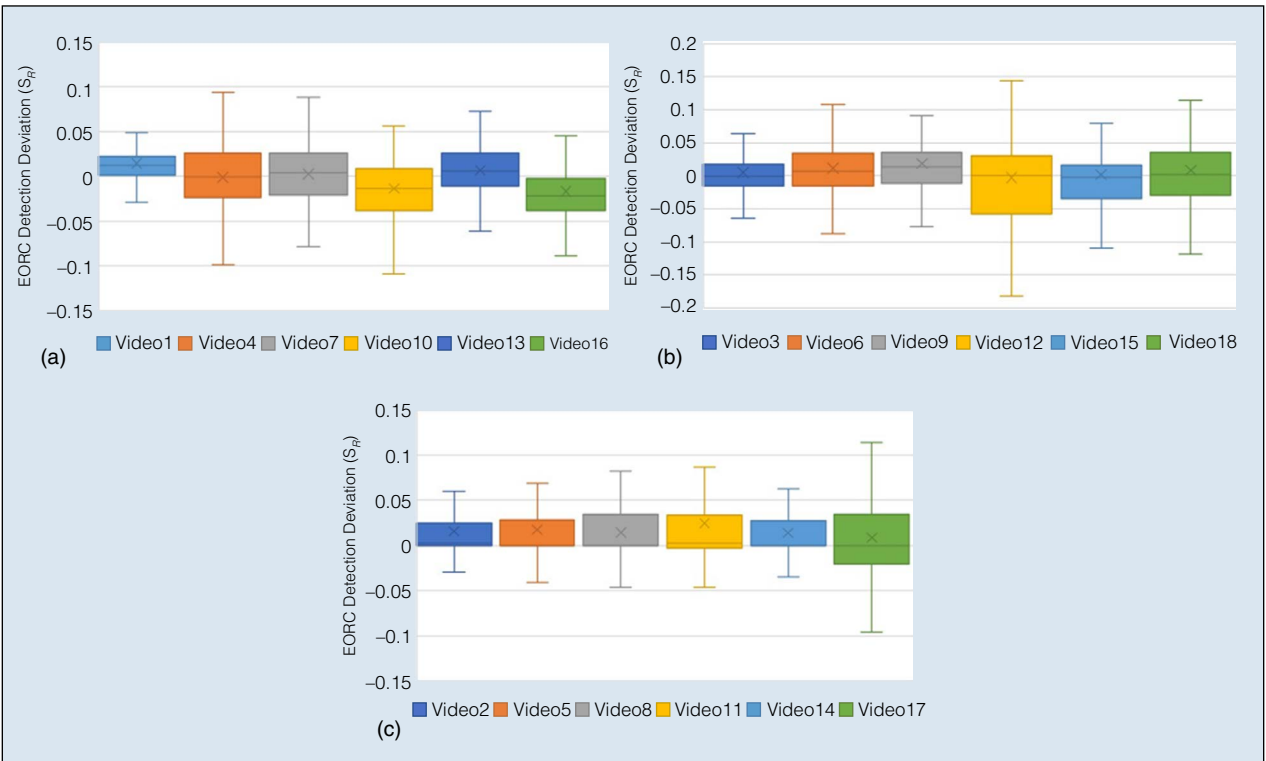


Fig. 6. Statistical boxplot results. (a) Non-fatigue state video; (b) Slight fatigue state video; (c) Severe fatigue state video.

that the current methods for establishing driver fatigue assessment models have diversity, but most of them are based on PERCLOS parameters [1],[2],[12] for research. The model proposed in this article compensates for the shortcomings of PERCLOS with a single parameter and further integrates ECD parameters to establish a Gaussian Mixture Model, in which the metrics of the eye status are described by the detected EOOCR value. The experiment results show that the EOOCR metrics have a direct relationship with fatigue level.

The presented work deals with the problem of driver fatigue detection in the field of traffic safety, which is a hot topic accompanied by some challenges and interesting possibilities presented by the latest technologies. The work has been framed to provide the reader with tutorial information about the theme as well as to highlight some new technical findings that characterize the proposal. Specifically, a novel detection method of driver multiple fatigue degree is described. Feature detection is based on improved iris segmentation and VGG network, and different metrics are output based on fatigue criteria based on Gaussian Mixture Model to subdivide different driver fatigue states. This work proposes the idea of subdividing fatigue degrees based on EOOCR metrics with the constructed Gaussian Mixture Model, which is different from the simple distinction between two states of fatigue or not, but can be subdivided into more fatigue states depending on the EOOCR metrics. The subdivision of multiple states is conducive to the fusion of other metrics for further research to improve the robustness of the system. The results of this method for driver fatigue degree identification are demonstrated through a short experiment case study.

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